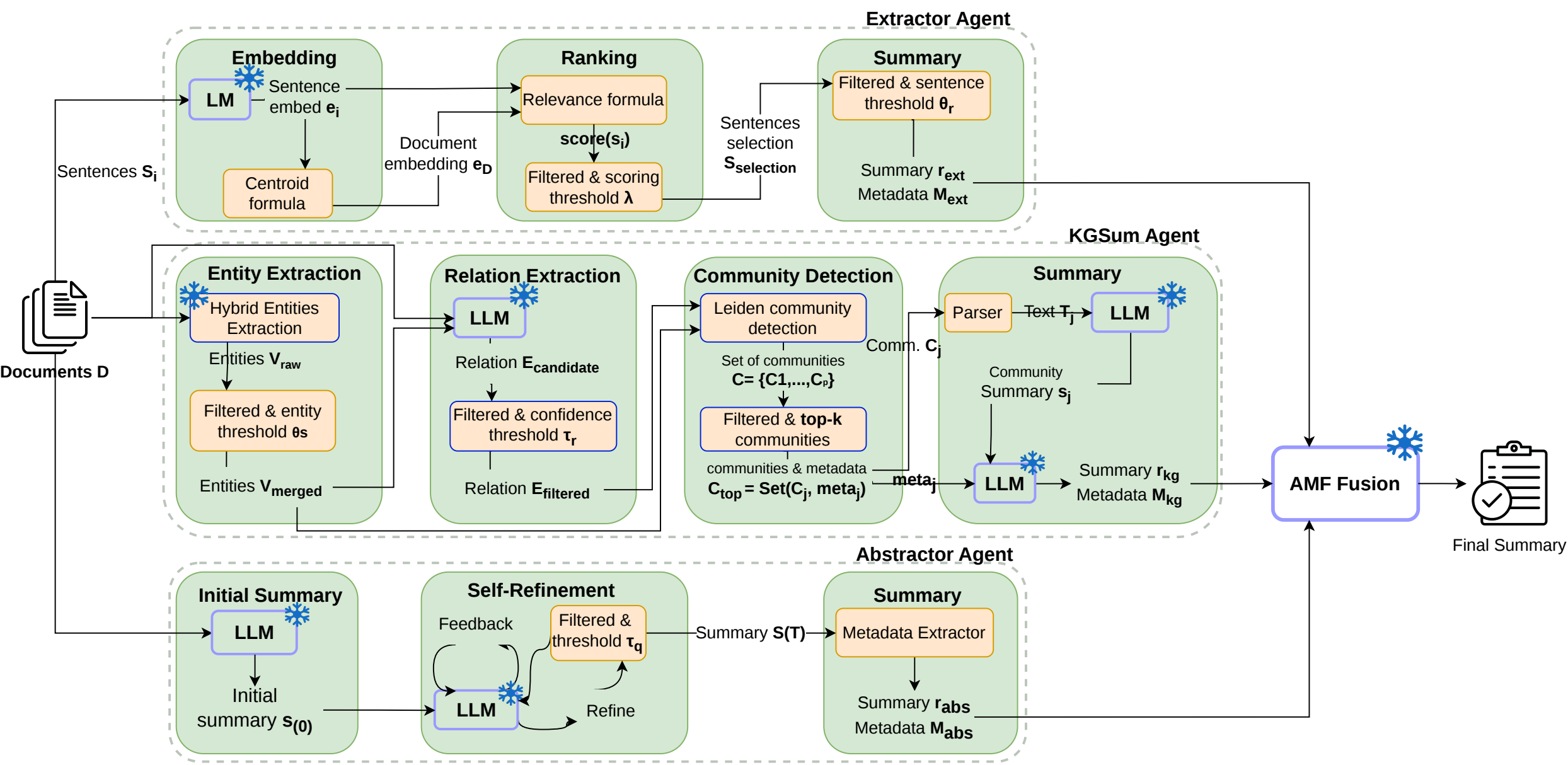

Extractor Agent
Embedding
LM
Sentence embed $e_i$
Centroid formula
Document embedding $e_D$
Ranking
Relevance formula
score($s_i$)
Filtered & scoring threshold λ
Sentences selection $S_{selection}$
Summary
Filtered & sentence threshold $\theta_r$
Summary $r_{ext}$
Metadata $M_{ext}$
Sentences $S_i$
KGSum Agent
Documents D
Entity Extraction
Hybrid Entities Extraction
Entities $V_{raw}$
Filtered & entity threshold θs
Entities $V_{merged}$
Relation Extraction
LLM
Relation $E_{candidate}$
Filtered & confidence threshold $\tau_r$
Relation $E_{filtered}$
Community Detection
Leiden community detection
Set of communities
C= {C1,...,Cp}
Filtered & top-k communities
communities & metadata
$C_{top}$ = Set($C_j$, $meta_j$)
Comm. $C_j$
$meta_j$
Summary
Parser
Text $T_j$
LLM
Community Summary $s_j$
LLM
Summary $r_{kg}$
Metadata $M_{kg}$
AMF Fusion
Final Summary
Abstractor Agent
Initial Summary
LLM
Initial summary $s_{(0)}$
Self-Refinement
Feedback
Filtered & threshold $\tau_q$
LLM
Refine
Summary S(T)
Summary
Metadata Extractor
Summary $r_{abs}$
Metadata $M_{abs}$